MILITARY SPECIFICATION

REPRODUCER, SIGNAL DATA: AN/GSQ-64
(READER, PUNCHED TAPE)

This amendment forms a part of Military Specification MIL-R-46332A(MU) dated 10 March 1965.

Paragraph 2.1, under SPECIFICATIONS: Add

"MIL-M-46336	Modules, Reproducer, Signal Data: AN/GSQ-64
MIL-T-60177	Transistor Assembly"

Paragraph 2.1, under DRAWINGS: Add

"B8623275	Tape, Test Set, Computer Logic Unit AN/GSM-70 (U) Self Test
D10531591	Terminal Board TB-6
F8213319	Amplifier Assembly, Photodiode
F10525770	Transistor Assembly
F10531686	Amplifier, Power Supply
F10531698	Signal Data Reproducer Subassembly
F10531712	Wiring Diagram - Signal Data Reproducer Subassembly
F10531828	Wiring Diagram Panel

Following paragraph 3.1, add paragraphs 3.1.1 and 3.1.2.

"3.1.1 List of assemblies.- The reproducer shall be comprised of the assemblies and modules listed herein. The assemblies and modules shall have been inspected and accepted in accordance with their respective drawings and specifications prior to incorporation in the reproducer.

"3.1.2 Continuity.- The reproducer shall exhibit continuity in accordance with wiring diagrams F10531712 and F10531828."

Assembly	Drawing	Specification
Terminal Board TB-6	D10531591	MIL-M-46336
Amplifier Assembly, Photodiode	F8213319	MIL-M-46336
Transistor Assembly	F10525770	MIL-T-60177
Amplifier, Power Supply	F10531686	MIL-M-46336
Amplifier, Electronic Control	F10531690	MIL-M-46336
Signal Data Reproducer, Subassembly	F10531698	MIL-M-46336

Paragraph 3.3.1: Delete and substitute new paragraph as follows:

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"3.3.1 Vibration.- The reproducer shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch maximum excurtion). The frequency shall be varied continuously from 10 to 55 cycles per second and return to 10 cycles and shall be transversed in approximately one minute ± 15 seconds. This motion shall be applied for a period of 10 minutes in each of the three mutually perpendicular direction (total of 30 minutes) at standard room ambient temperature (60°F to 90°F). Subsequent to vibration the reproducer shall not have been damaged or adversely affected and shall meet the electrical performance requirements of this specification."

Paragraph 3.3.2: Delete "15 gravity units" and substitute "5" gravity units".

Paragraph 3.4.3.1: Change "10.0 mv" to 20.0 mv".

* Paragraph 3.4.3.4: Delete in its entirety.

Paragraph 3.5.1: Delete "52 inches" and substitute "44.5 inches". Delete "520 frames" and substitute "445 frames".

Paragraph 3.5.4: Delete in its entirety and substitute the following:

"3.5.4 Diameter and spacing of punched holes.- The diameter and spacing of punched holes shall be in accordance with Drawing B8623275."

Table I, Column 5 under (Max): Delete "15.6" and substitute "15.8". Column 9 under (Max): Delete "15.6" and substitute "15.85".

Paragraph 3.6(a), delete first sentence and substitute the following:

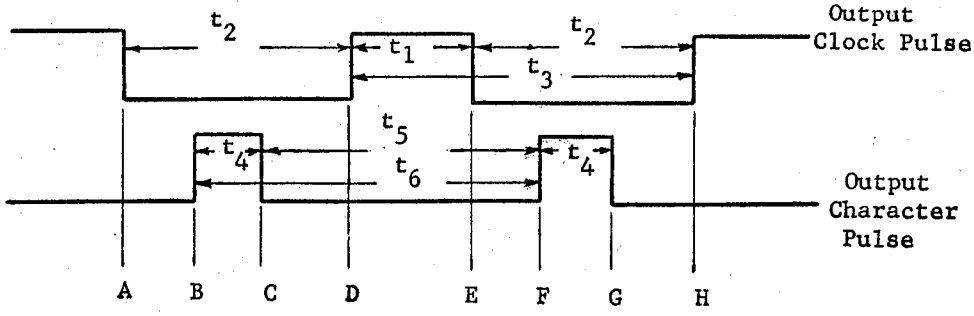
"The bias voltage with respect to ground, measured at the terminal board connector pin listed in 3.6(c), shall have a minimum range of zero to minus 1.1 volts as the associated potentiometer is adjusted."

* Paragraph 3.6(c) Figure 1, change "-0.8 ± 0.2 V" to "-0.8 ± 0.2V" 0.6V"

* Paragraph 3.6.1: Change third, fourth, fifth, sixth and seventh sentence as follows:

"Pulse duration for t1 + t2 shall be 2.25 msec ± 10%. The normal pulse duration for t5 shall be 1.54 ± .05 msec. At temperature extremes the pulse duration for t5 shall be 1.54 ± .1 msec." Fourth sentence: delete "between the minimum and maximum values specified in Table II for the appropriate ambient temperature," and substitute "more negative than - 5.5 volts". Fifth and sixth sentences: change "5,600 ohm to 600 ohm". Seventh sentence: Change "The peak amplitude of the positive going pulse" to "The peak amplitude of the negative going pulse.

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Output Clock and Character Pulses

1. DE = t_1; AD = EH = t_2
2. DH = $t_1 + t_2 = t_3$ = One Clock Cycle
3. BC = FG = t_4; CF = t_5
4. BF = $t_4 + t_5 = t_6$ = One Character Cycle
5. DH = BF = 2.25 msec $\pm$ 10% = One Cycle
6. CF = t_5 = 1.54 $\pm$ .05 msec at room ambient
7. When clock and character pulses are displayed on a dual trace oscilloscope, the distance AB and/or EF shall be greater than 8% of one cycle.

FIGURE 2

* Paragraph 3.6.2, delete paragraph 3.6.2 and substitute:

"3.6.2 Strobe pulses.- With the tape advance switch in OFF position, energizing the reproducer and electrically connecting Pin W of output connector PB to Pin 13 of J10 shall advance the tape. The reproducer shall simultaneously deliver an oscilloscope presentation of the strobe (clock) pulse shown in Figure 2. Nominal pulses duration (t_1 shall be $35 \pm {}^{15}_{0}$ percent of the entire 2.25 msec $\pm$ 10% cycle ($t_1 + t_2$) shown in Figure 2. Under the above conditions the peak pulses amplitude shall be as indicated in last line, last six columns of Table II. This pulse shall be obtained across a 5,600 ohm resistive load, existing between reproducer output connector Pin S and power supply ground, Pin N, and at a point within the range of rotation of potentiometer R9. The pulse duration for t_1 shall be 35 $\pm$ 15 percent at the temperature extremes."

Table II: Delete the last line under the last six columns and substitute the following:

" $\frac{VR}{11300 + R}$	$\frac{VR}{8700 + R}$	$\frac{VR}{10500 + R}$	$\frac{VR}{9500 + R}$	$\frac{VR}{10700 + R}$	$\frac{VR}{9300 + R}$

V = Actual 15.0 volt level at temperature test.
R = Actual Value of 5600 ohm resistor at time of temperature test."

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* Delete paragraph 3.6.3 in its entirety and substitute the following:

"With the reproducer energized and the TAPE ADVANCE switch in the OFF position (tape stopped), the voltage measured across a 600 ohm load connected between pin M and pin N (power supply ground) of connector PB, shall be more positive than -0.5 VDC. When the TAPE ADVANCE switch is placed momentarily to ON position and back to OFF position (Tape running), the voltage across the 600 ohm load shall be more negative than -7.0 VDC. The above readings shall be obtained at the temperatures extremes."

Paragraph 3.7: Delete "3.6a and".

* 3.6.8 Delete in its entirety and substitute:

"When the reproducer is energized and the FIELD DATA TELETYPE SWITCH is placed in the Teletype position, the positive voltage between reproducer output connector PB, Pin V and power supply ground, Pin N, should be as specified in Level 5 of Table I for each indicated ambient temperature. The above voltages shall be obtained when Pin C of output connector PB is electrically connected to Pin 5 of J10."

* Paragraph 3.6.12. Change "16" to "20," and add to sentence, "18 to 21.7VDC"

Following paragraph 4.1.1, add paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2.

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) reproducers manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same course of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

* Table III: Line 6: Delete in its entirety. Add the following to the columns as shown:

"Characteristic	Requirements	Test Procedures
Continuity	3.1.2	4.5.2(a)"

Paragraph 4.5.1.1: Delete "inspected" and substitute "tested in last sentence; delete "4.5.1.2" and substitute "4.5.2 to 4.5.4.14" in last sentence.

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Paragraph 4.5.1.2: Delete "inspected" and substitute "tested" in last sentence; delete "4.5.1.3" and substitute "4.5.2 to 4.5.4.12" in last sentence.

Paragraph 4.5.1.3, in last sentence, delete "4.5.1.4" and substitute "4.5.4.14.",

Following paragraph 4.5.2: Add new paragraph 4.5.2(a):

"4.5.2(a) Continuity.- The point to point wiring shall be tested with a conventional type tester for compliance with 3.1.2 prior to performing the tests specified in 4.5 through 4.5.4.14.

Paragraph 4.5.2.3, Table VI, delete "10.0 millivolts" and substitute "20.0 millivolts".

* Table VI: Delete line 4 in its entirety.

Paragraph 4.5.2.4, delete paragraph 4.5.2.4 and substitute:

"4.5.2.4 Homing.- Set power and signal switches to the "OFF" position Set FILL-Verify switch in FILL position. Load Tape II (Item 6 of 4.4.1.1) into the read head. Start the tape by placing the POWER and SIGNAL switches to "ON" position and by placing the tape ADVANCE switch from OFF position momentarily to the ON position and back to the OFF position. The tape advance shall be stopped when the homing slot is positioned in the photo diode read head. Set the FILL-Verify switch to the VERIFY position, restart the tape as noted above, observing that the tape advances continuously. After meeting this test at the standard ambient temperature (60°F to 90°F) in compliance with 3.4.4 the test shall be repeated at minus 40°F and plus 125°F."

Paragraph 4.5.2, delete 4.4.1.1.

Paragraph 4.5.2.1: Change "156 inches" to "133.5 inches".

* Paragraph 4.5.3.2: On the fifth line change "15-25 percent" to "15-30 percent". Add to last sentence: "The pulse duration for t_1 shall be 35 ± 15% at temperature extremes". On the sixth line change "Negative" to"Positive".

Paragraph 4.5.4.1, delete last sentence and substitute "This test is only to be performed during initial adjustment and when requirements of 4.5.4.2 and 4.5.4.3 are not met and readjustment is necessary."

Paragraph 4.5.4.2, delete last sentence.

Paragraph 4.5.4.3, delete"80 ± 5% of the entire cycle" and substitute "1.54 ± 0.05 msec at ambient (60°F to 90°F) and 1.54 $^{+0.1}_{-0.3}$ msec at the extreme temperatures (-40°F and +125°F)."

Paragraph 4.5.4.4: Delete "shall measure between minus 14.0 and minus 16.0 volts" and substitute "across the 5,600 ohm load at Pins S and N shall be as specified in 3.6"; delete "and the values specified in Table II for these temperatures."

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* Paragraph 4.5.4.5: Delete in its entirety and substitute the following:

"4.5.4.5 Functional Tr output.- The Tape Advance Switch shall be placed in the "OFF" position. Connect a 600 ohm load (1/2 watt, 5%) between Pin M and Pin N (Power Supply ground) of Connector PB. A multimeter, Item 1, shall be connected between Pin M and Pin N. The measured voltage shall be more positive than minus 0.5 VDC. The Tape Advance Switch shall be momentarily placed in the "ON" position. Then place the Tape Advance Switch to the "OFF" position. The measured voltage shall be more negative than minus 7.0 VDC. After meeting this test at the standard ambient temperature, the test shall be repeated at minus 40°F and plus 125°F."

Paragraph 4.5.5: Delete "4.5.4.1 and".

* Paragraph 4.5.4.10: In second sentence change "Pin 14 of J10" to Pin 5 of J10".

*The outside margins have been marked to indicate where additions to be previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous amendment.

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